

A Uniform Linear-Depth Profile for the Upper Counterloop Band

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Abstract

We resolve the deterministic graded-counterloop budget at every integer depth below the second alias. Let $y_k = (\beta_k R)^2$, $x = (\beta R)^2 > 1$, and $y_k = x \exp[-\log C_M/k + o(k^{-1})]$. If

$$A_k = (1 - k^{-1})y_k^k, \quad B_k(L) = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}, \quad 1 \leq L \leq k-1,$$

then, uniformly over the complete range of L ,

$$B_k(L) = \frac{y_k^{k+L+1}(1 - y_k^{-L})}{(k+L)(y_k - 1)} (1 + O(k^{-1})).$$

The source-locked form and the ratio to the first alias are

$$B_k(L) = \frac{x^{k+L+1}(1 - x^{-L})}{C_M^{1+L/k}(k+L)(x-1)}(1 + o(1)), \quad \frac{B_k(L)}{A_k} = \frac{x^{L+1}(1 - x^{-L})}{C_M^{L/k}(k+L)(x-1)}(1 + o(1)),$$

again with uniform relative errors. Consequently, if $L/k \rightarrow \alpha \in (0, 1]$, the constants $C_M^{-(1+\alpha)}$ and $C_M^{-\alpha}$ and the denominator factor $(1 + \alpha)^{-1}$ are unavoidable; the k th-root rates are $x^{1+\alpha}$ and x^α . For $L = \lfloor \alpha k + c \rfloor$, the phase $\theta_k = \{\alpha k + c\}$ remains in the normalized leading term. Rational slopes give a finite periodic orbit, while irrational slopes give the full closed phase interval. The actual-head transfer is stated only conditionally on the original unnormalized same-clock leaf $D_{4k}(R) \rightarrow 0$. No root or rank identification, Gate A–E promotion, Hilbert–Polya construction, zero identification, or proof of RH is claimed.

1 Source-locked counterloop ledger

Fix the physical constants

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \frac{28}{17} < \lambda < \frac{17}{10}. \quad (1)$$

Let $k = k_\sigma \rightarrow \infty$ be the first-alias clock used in RH-342. The boundary multiplier law from RH-17 is

$$|M_k| = C_M \lambda^k \{1 + o(1)\}, \quad C_M > 0. \quad (2)$$

The Hardy-normalized radii are

$$\beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}, \quad (3)$$

so that

$$x = (\beta R)^2 = \frac{(R/r_H)^2}{\lambda} > 1, \quad y_k = (\beta_k R)^2 = x \exp \left[-\frac{\log C_M}{k} + o(k^{-1}) \right]. \quad (4)$$

The strict inequality $x > 1$ follows from the certified upper bound on $\lambda[3, 4]$.

The finite graded counterloop is

$$\mathcal{Y}_k = \{\beta_k e^{ij\pi/k}, \beta_k e^{-ij\pi/k} : 1 \leq j \leq k-1\}. \quad (5)$$

Its exact power ledger is

$$s_{k,n} = \sum_{\nu \in \mathcal{Y}_k} \nu^n = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n) \quad (6)$$

$[1, 2]$. Thus the first alias at order $2k$ has weighted modulus

$$A_k := \frac{|s_{k,2k}| R^{2k}}{2k} = (1 - k^{-1}) y_k^k. \quad (7)$$

For $1 \leq L \leq k-1$, define the even post-first-alias budget

$$B_k(L) := \sum_{j=1}^L \frac{|s_{k,2k+2j}| R^{2k+2j}}{2k+2j} = \sum_{j=1}^L \frac{y_k^{k+j}}{k+j}. \quad (8)$$

The upper bound $L \leq k-1$ is exactly the strict band $2k < 2k+2L < 4k$; the odd counterloop moments in this band vanish.

Proposition 1.1 (Exact ratio). *For every finite $k \geq 2$ and $1 \leq L \leq k-1$,*

$$\frac{B_k(L)}{A_k} = \frac{k}{k-1} \sum_{j=1}^L \frac{y_k^j}{k+j}. \quad (9)$$

Proof. In the strict upper band, (6) gives $s_{k,2k+2j} = -2\beta_k^{2k+2j}$. Substitution into (8) and division by (7) gives (9). $\square$

2 Uniform endpoint localization

Theorem 2.1 (Complete-band endpoint theorem). *Assume (4). There is a constant (depending only on the eventual compact neighborhood of x) such that, uniformly for every integer $1 \leq L \leq k-1$,*

$$B_k(L) = \frac{y_k^{k+L+1}(1 - y_k^{-L})}{(k+L)(y_k - 1)} \left(1 + O(k^{-1})\right). \quad (10)$$

Moreover, uniformly over the same complete band,

$$B_k(L) = \frac{x^{k+L+1}(1 - x^{-L})}{C_M^{1+L/k}(k+L)(x-1)} (1 + o(1)), \quad (11)$$

$$\frac{B_k(L)}{A_k} = \frac{x^{L+1}(1 - x^{-L})}{C_M^{L/k}(k+L)(x-1)} (1 + o(1)). \quad (12)$$

All $o(1)$ terms in this theorem are relative errors uniform in L .

Proof. Reindex the sum in (8) from its terminal coordinate:

$$B_k(L) = \frac{y_k^{k+L}}{k+L} \sum_{r=0}^{L-1} y_k^{-r} \frac{k+L}{k+L-r}. \quad (13)$$

Since $y_k \rightarrow x > 1$, choose $1 < \xi < x$ so that $y_k \geq \xi$ for all large k . Writing the last factor as $1 + r/(k + L - r)$ gives

$$0 \leq \sum_{r=0}^{L-1} y_k^{-r} \left(\frac{k+L}{k+L-r} - 1 \right) \leq \frac{1}{k+1} \sum_{r \geq 0} r \xi^{-r} = O(k^{-1}).$$

The unweighted geometric sum is at least 1, so the sum in (13) equals

$$\sum_{r=0}^{L-1} y_k^{-r} \{1 + O(k^{-1})\} = \frac{y_k}{y_k - 1} (1 - y_k^{-L}) \{1 + O(k^{-1})\},$$

with the relative error uniform in L . This proves (10).

For the source-locked replacement, put

$$\rho_k := k \log(y_k/x) + \log C_M \rightarrow 0.$$

Then, uniformly for $L \leq k - 1$,

$$\frac{y_k^{k+L+1}}{x^{k+L+1} C_M^{-(1+L/k)}} = C_M^{-1/k} \exp((1 + L/k + 1/k) \rho_k) \rightarrow 1. \quad (14)$$

The function

$$G_L(t) := \frac{1 - t^{-L}}{t - 1} = \sum_{r=1}^L t^{-r}$$

has $|G'_L(t)| \leq \sum_{r \geq 1} r \xi^{-r-1}$ for $t \geq \xi$ and $G_L(x) \geq x^{-1}$. Hence $G_L(y_k)/G_L(x) \rightarrow 1$ uniformly in $L \geq 1$. Combining this observation with (10) and (14) proves (11). Finally,

$$A_k = (1 - k^{-1}) y_k^k = C_M^{-1} x^k (1 + o(1)),$$

and division of (11) by this expression gives (12); the factor C_M^{-1} cancels from the ratio. $\square$

Remark 2.2 (Why endpoint localization is uniform). The denominator is not replaced by k when L is linear. Instead, after reversal, the geometrically decaying weight y_k^{-r} makes the weighted average of r bounded, while every denominator $k + L - r$ is at least $k + 1$. This is the reason the complete-band error remains $O(k^{-1})$.

3 Linear depth and physical-clock rates

Corollary 3.1 (Linear-depth profile). *If $L = L_k$ satisfies $1 \leq L_k \leq k - 1$ and*

$$\frac{L_k}{k} \rightarrow \alpha \in (0, 1],$$

then

$$A_k \sim C_M^{-1} x^k, \quad (15)$$

$$B_k(L_k) \sim \frac{x^{k+L_k+1}}{C_M^{1+\alpha} k(1+\alpha)(x-1)}, \quad (16)$$

$$\frac{B_k(L_k)}{A_k} \sim \frac{x^{L_k+1}}{C_M^\alpha k(1+\alpha)(x-1)}. \quad (17)$$

In particular,

$$\lim_{k \rightarrow \infty} B_k(L_k)^{1/k} = x^{1+\alpha}, \quad \lim_{k \rightarrow \infty} \left(\frac{B_k(L_k)}{A_k} \right)^{1/k} = x^\alpha. \quad (18)$$

Proof. Because $\alpha > 0$, $L_k \rightarrow \infty$, so $1 - x^{-L_k} \rightarrow 1$. Also $k + L_k \sim (1 + \alpha)k$ and $C_M^{L_k/k} \rightarrow C_M^\alpha$. Substitution into theorem 2.1 gives (16) and (17); (15) follows directly from (7) and (4). Taking logarithms and dividing by k gives (18); all polynomial factors and fixed positive constants have zero logarithmic rate. $\square$

On the physical first-alias clock, RH-355 records

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1). \quad (19)$$

Thus the terminal order $n_k = 2k + 2L_k$ obeys

$$\frac{n_k}{\log(1/\sigma)} \rightarrow \frac{1 + \alpha}{\log \lambda}, \quad \frac{n_k - 2k}{\log(1/\sigma)} \rightarrow \frac{\alpha}{\log \lambda}. \quad (20)$$

The budget logarithmic rates are

$$\frac{\log B_k(L_k)}{\log(1/\sigma)} \rightarrow \frac{(1 + \alpha) \log x}{2 \log \lambda}, \quad (21)$$

$$\frac{\log(B_k(L_k)/A_k)}{\log(1/\sigma)} \rightarrow \frac{\alpha \log x}{2 \log \lambda}. \quad (22)$$

These are deterministic counterloop rates, not probabilities or actual-head measurements.

4 Phase-safe integer normalization

Theorem 4.1 (Floor phase at linear depth). *Fix $\alpha \in (0, 1]$ and $c \in \mathbb{R}$, and let*

$$L_k = \lfloor \alpha k + c \rfloor, \quad \theta_k = \{\alpha k + c\}.$$

Assume $1 \leq L_k \leq k - 1$ eventually (for $\alpha = 1$ this requires $c < 0$). Then

$$kx^{-(1+\alpha)k} B_k(L_k) = \frac{x^{c+1-\theta_k}}{C_M^{1+\alpha}(1+\alpha)(x-1)}(1 + o(1)), \quad (23)$$

$$kx^{-\alpha k} \frac{B_k(L_k)}{A_k} = \frac{x^{c+1-\theta_k}}{C_M^\alpha(1+\alpha)(x-1)}(1 + o(1)). \quad (24)$$

If $\alpha = a/b$ is rational in lowest terms, θ_k has the exact period- b orbit

$$\left\{ \left\{ \frac{ar}{b} + c \right\} : 0 \leq r < b \right\};$$

the normalized quantities have the corresponding finite cluster sets. If α is irrational, $\{\theta_k : k \geq 1\}$ is dense in $[0, 1]$, and the closed cluster sets in (23) and (24) are, respectively,

$$\left[\frac{x^c}{C_M^{1+\alpha}(1+\alpha)(x-1)}, \frac{x^{c+1}}{C_M^{1+\alpha}(1+\alpha)(x-1)} \right], \quad (25)$$

$$\left[\frac{x^c}{C_M^\alpha(1+\alpha)(x-1)}, \frac{x^{c+1}}{C_M^\alpha(1+\alpha)(x-1)} \right]. \quad (26)$$

Proof. Write $L_k = \alpha k + c - \theta_k$ in (16) and (17); the two displayed normalizations follow directly. For rational a/b , adding a/b permutes the b residue classes modulo one, which gives the stated finite orbit. For irrational α , the standard irrational-rotation lemma says that the fractional parts of $k\alpha + c$ are dense modulo one. A short proof uses the pigeonhole approximation $0 < \|q\alpha\| < 1/N$ and the translates of the resulting fine mesh; letting $N \rightarrow \infty$ gives a point in every subinterval of $[0, 1]$. The continuous map $t \mapsto x^{c+1-t}$ then maps the closed phase set to the intervals in (25)–(26). $\square$

Remark 4.2 (The RH-355 endpoint). For $\alpha = 1$ and any $c \in [-1, 0)$, $L_k = k - 1$ and $\theta_k = c + 1$. Equation (23) becomes

$$kx^{-2k}B_k(k-1) \longrightarrow \frac{1}{2C_M^2(x-1)},$$

which is exactly the complete strict upper-band constant of RH-355. The phase notation is still useful for terminal lags other than one.

5 Boundary stitching and conditional actual-head transfer

The complete theorem includes bounded depths, but the simplification $1 - x^{-L} \sim 1$ does not. If L is fixed, the factor is a leading constant. If $L \rightarrow \infty$ while $L = o(k)$, it may be deleted and RH-356's mesoscopic law is recovered. The case $L/k \rightarrow 0$ with bounded L is therefore not the same asymptotic regime as the case $L/k \rightarrow 0$ with $L \rightarrow \infty$.

At the other endpoint $L = k - 1$, (8) is the whole strict upper band and the theorem recovers RH-355. No interpolation of these deterministic budgets is an actual-head result.

Let $\mathcal{H}_\sigma$ be the actual modulus-complete Hardy head of RH-342, with moments $h_{\sigma,n}$, and put

$$d_{\sigma,k,n} = h_{\sigma,n} - s_{k,n}.$$

The original unnormalized same-clock transport leaf is

$$D_{4k}(R) := \sum_{2 \leq n < 4k} \frac{|d_{\sigma,k,n}|R^n}{n} \longrightarrow 0. \quad (27)$$

It is an open hypothesis, not proved in this paper. Define

$$A_k^{\mathcal{H}} := \frac{|h_{\sigma,2k}|R^{2k}}{2k}, \quad (28)$$

$$B_k^{\mathcal{H}}(L) := \sum_{j=1}^L \frac{|h_{\sigma,2k+2j}|R^{2k+2j}}{2k+2j}, \quad (29)$$

$$O_k^{\mathcal{H}}(L) := \sum_{2k < n \leq 2k+2L-1, n \text{ odd}} \frac{|h_{\sigma,n}|R^n}{n}. \quad (30)$$

Theorem 5.1 (Conditional uniform inheritance). *Assume (27) on one physical clock. Then*

$$\frac{A_k^{\mathcal{H}}}{A_k} \rightarrow 1, \quad \sup_{1 \leq L \leq k-1} \left| \frac{B_k^{\mathcal{H}}(L)}{B_k(L)} - 1 \right| \rightarrow 0, \quad \sup_{1 \leq L \leq k-1} O_k^{\mathcal{H}}(L) \rightarrow 0. \quad (31)$$

Consequently the linear-depth and phase-safe budget formulas transfer to $B_k^{\mathcal{H}}(L)/A_k^{\mathcal{H}}$ only under this hypothesis.

Proof. The reverse triangle inequality and the fact that every displayed order lies below $4k$ give

$$|A_k^{\mathcal{H}} - A_k| \leq D_{4k}(R), \quad |B_k^{\mathcal{H}}(L) - B_k(L)| \leq D_{4k}(R).$$

Since $A_k \sim C_M^{-1}x^k$ and, uniformly for $L \geq 1$,

$$B_k(L) \geq B_k(1) = \frac{y_k^{k+1}}{k+1} \sim \frac{x^{k+1}}{C_M k},$$

division by these diverging lower bounds proves the first two assertions. The counterloop odd moments vanish, so each actual odd budget is bounded by $D_{4k}(R)$. The ratio assertion follows by dividing the two inherited even budget asymptotics. This is a conditional budget statement only; it neither identifies actual roots or rank nor proves (27). $\square$

6 Scope, next route, and reproducibility

The endpoint theorem is a deterministic statement about the source-locked finite graded shell. Its numerical rows are exact rational or high-precision synthetic checks and are not finite-order evidence for an all-order theorem beyond the proof above. In particular, the paper does not replace the counterloop by an actual noisy operator, turn a local law into a trace law, or identify any Riemann zero.

The next read-only candidate is RH-358: reindex the upper band by the terminal lag $q = k - 1 - L$ and quantify the resulting geometric localization. It should remain on the deterministic counterloop until a typed actual-head theorem is available.

Reproduction. The accompanying Python artifact implements exact rational rows, the full-band endpoint-error envelope, rational phase orbits, and high-precision synthetic linear-depth rows. It explicitly marks every row as formula reproduction only. Build and archive commands are listed in the README.

References

- [1] Bin Wang. The one- and two-alias ledger at the natural counterloop clock, 2026. Repository paper RH-297.
- [2] Bin Wang. A common-hardy head/counterloop rank lock and a hidden-shell obstruction, 2026. Repository paper RH-342.
- [3] Bin Wang. Projector-mass first-alias threshold and an isospectral cell obstruction, 2026. Repository paper RH-336.
- [4] Liang Wang. Time-ordered boundary monodromy at a quadratic band-merging map, 2026. Repository paper RH-17.